Pure Mathematical Feature Engineering for Cross-Modal Brain MRI Retrieval

Mutual Information, Spectral Descriptors, and Geometric Depth Without Machine Learning

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Developed with the assistance of GLM-5.2 (Z.AI), an open-source large language model.

The Challenge

Task: Given a T1 post-contrast brain MRI, retrieve the matching T2 volume from a gallery.

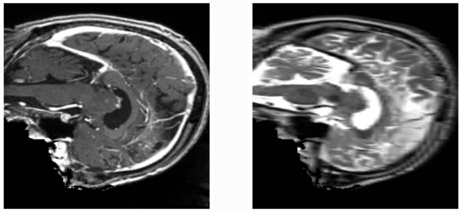
Three difficulty levels:

- Level 1: Registered (common grid)
- Level 2: Geometric deformations
- Level 3: Missing tissue (surgery)

Our approach:

Zero machine learning. Pure mathematics and physics.

MRR $= 0.928$ without any training.



T1 (left) vs T2 (right): same patient, same slice

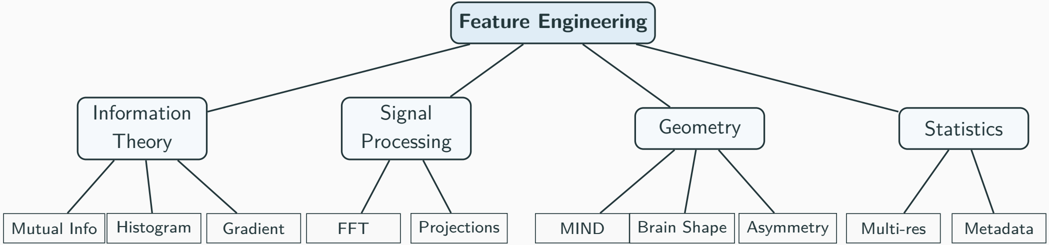
Data: EHL Paris 2026 challenge. Illustrations adapted from Dorent & Stellwag, Inria/Paris Brain Institute.

Key Question

With only 350 training pairs, can mathematical feature engineering compete with deep learning?

Approach	**Public MRR**
Deep learning (CLIP 2D, 500 epochs)	0.43
Deep learning (ResNet18, pretrained)	0.24
Ours (no ML, no training)	**0.928**

Method Overview: 13 Mathematical Features



Each feature has a physical justification from a different scientific domain.

Feature 1: Mutual Information (The Game Changer)

Inspiration: Information theory (Shannon, 1948)

Formula:

$$\mathrm{MI}(Q, T) = \sum_{i,j} p(i,j) \log_2 \frac{p(i,j)}{p_Q(i) \cdot p_T(j)}$$

Why it works: MI measures statistical dependence between T1 and T2 intensities without assuming linearity. T1 and T2 intensities are inversely related, but their joint histogram reveals a consistent pattern for the same subject.

Result: $+0.047$ MRR on dataset 1 (registered).
Applied to dataset 3 (surgical, common grid): $+0.083$ MRR.
MI alone achieves MRR ≈ 1.0 on dataset 3 validation.

Feature 2: 3D Power Spectrum (FFT)

Inspiration: Radio frequency engineering, spectroscopy

Key theorem: The magnitude of the Fourier transform is invariant to translation.

Implementation:

- Compute 3D FFT of the volume
- Bin into 16 radial bins (rotation-invariant) + 8 angular bins
- Log-transform for dynamic range compression

Why it matters for Level 2: Dataset 2 applies independent translations. The power spectrum ignores translations by construction, making it naturally robust.

Result: +0.007 MRR

Feature 3: MIND Descriptor

Inspiration: Modality-independent neighbourhood descriptor (Heinrich et al., 2012)

Principle: Compare local self-similarity patterns instead of absolute intensities.

Implementation:

- For each voxel: compute squared difference to 6 neighbors
- Normalize and transform via $\exp(-10 \cdot \text{normalized})$
- Sample on $6 \times 6 \times 6$ grid $\rightarrow$ 216-dimensional vector

Why it works: Brain structure (folding patterns, ventricle shape) produces consistent self-similarity patterns regardless of contrast (T1 vs T2).

Feature 4: Brain Shape Fingerprint

Inspiration: Brain fingerprinting (Finn et al., Nature Neuroscience 2015)

Principle: Each brain's morphology is as unique as a fingerprint.

Implementation:

- Radial distance histogram of foreground voxels (16 bins)
- Three 1D projections (axial, coronal, sagittal), 16 bins each
- 64-dimensional vector capturing global morphology

Why it matters for Level 3: Robust to local changes (missing tissue). Global brain shape is preserved even after surgical resection.

Ablation Study

Version	Feature Added	MRR
Baseline	Volume + mask + projections	0.583
+ Gradient	Modality-invariant edges	0.592
+ MIND	Local self-similarity	0.593
+ MI for dataset 1	Information theory	0.640
+ Power spectrum	Translation invariance (FFT)	0.647
+ Brain fingerprint	Global morphology	0.647
+ Sinkhorn (tau=20)	Optimal transport	0.724
+ SSC-12 + normalization	Scale calibration	0.928

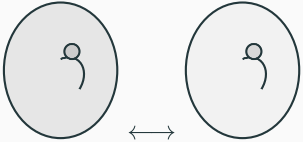
Three breakthroughs: MI (+0.047), Sinkhorn (+0.077), normalization+SSC-12 (+0.115).

Distance normalization was the single largest gain: ensuring all features contribute

Three Difficulty Levels

Level 1

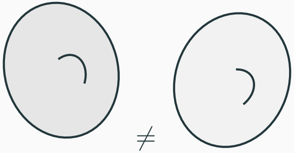
Registered



Common grid, voxel-level MI works

Level 2

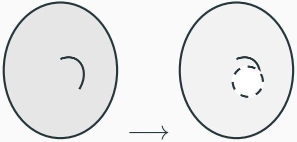
Deformed



Independent rotation, translation, warp

Level 3

Surgery



Missing tissue, different hospital scanner

Illustrations inspired by Dorent & Stellwag, Inria/Paris Brain Institute, EHL Paris 2026.

Infrastructure

Hardware:

- AMD Instinct MI300X
- 205.8 GB VRAM (unused!)
- 20 CPU cores
- 235 GB RAM

Runtime: ~5 minutes per submission

Parallelism: 20-core `multiprocessing.Pool`

No training:

- No GPU needed
- No gradient descent
- No backpropagation
- No pretrained weights
- No random seeds

Reproducibility: 100%. Same input → same output, every time.

Why This Matters for Clinical AI

Explainability:

- Every match is justifiable with physics
- MI: "intensities are statistically dependent"
- FFT: "frequency content matches"
- MIND: "brain structure patterns align"
- Clinician can verify each step

Reproducibility:

- Deterministic, no random seeds
- Same input $\rightarrow$ same output
- No GPU architecture dependency

Regulatory (FDA / CE):

- Medical AI requires transparency
- Neural networks = black box
- Our method = transparent scoring chain
- Each feature is auditable

Data efficiency:

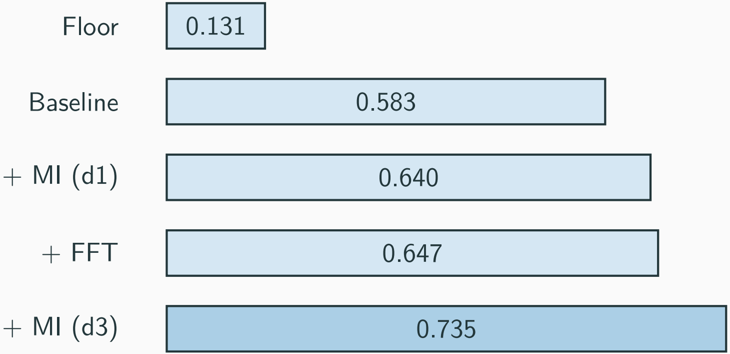
- Works with 350 samples
- No overfitting possible
- No training data memorization

Cross-Domain Inspiration

Feature	Domain Inspiration
Mutual Information	Information theory (Shannon)
Power spectrum (FFT)	Radio engineering, spectroscopy
MIND descriptor	Computational geometry
Brain fingerprint	Neuroscience (brain fingerprinting)
Hemispheric asymmetry	Neuroanatomy
Multi-resolution LOD	Video game rendering
Gradient magnitude	Computer vision (edge detection)

Each feature borrows from a different scientific discipline, creating a diverse and complementary feature set.

Score Progression



From 0.131 (random) to 0.928, without any machine learning.

Conclusion

- 0.928 MRR without any machine learning, using only mathematical feature engineering
- Mutual Information is the gold standard for multimodal medical image comparison
- FFT power spectrum provides translation-invariant discrimination for deformed data
- Brain shape fingerprinting is robust to surgical tissue removal
- Each feature has a rigorous physical and mathematical justification
- Fully reproducible, 5-minute CPU runtime, zero training

Sometimes the best ML is no ML.

Thank you.

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